EXERCÍCIOS NÚMEROS COMPLEXOS

RESOLVIDOS POR: GABRIEL E.S

$x_1 = 2 + j3$, $x_2 = 4 - j3$, $x_3 = 5 \angle 0{,}927$ → $x_3 = 3 + 4i$

↳ $a = 5 \cdot \cos(0{,}927)$

$a \cong 3$

$b = 5 \cdot \operatorname{sen}(0{,}927)$

$b = 4$

a) $x_1 + x_2$

$2 + \cancel{3j} + 4 - \cancel{3j}$

$6$

b) $x_2 + x_3$

$4 - 3j + 3 + 4j$

$= 7 + j$

c) $x_2 - x_3$

$4 - 3j - (3 + 4j)$

$4 - 3j - 3 - 4j$

$1 - 7j$

D) $x_2 \cdot x_3$

$(4 - 3j) \cdot (3 + 4j)$

$12 + 16i - 9j - 12j^2$

$12 + 7j + 12$

$24 + 7j$

$j^2 = \sqrt{-1}^2 = -1$

e) $x_2 / x_3$

$$\frac{4 - 3j}{3 + 4j} \cdot \left(\frac{3 - 4j}{3 - 4j}\right)$$

$$\frac{\cancel{12} - 16j - 9j + \cancel{12j^2}}{9 - 16j^2} = -\frac{25j}{+25} = -j$$

$j^2 = -1$

f) $Re(x_3) = 3$

g) $Im(x_3) = 4j$

h) $|x_1| = \sqrt{a^2 + b^2}$

$a = 2$

$b = 3$

$= \sqrt{2^2 + 3^2}$

$= \sqrt{13}$

2) a) $(6 + 7j) \cdot (1 + j)$ $\quad j^2 = -1$

$= 6 + 6j + 7j + 7\cancel{j^2}^{-1}$

$= -1 + 13j$

b) $(5+j4)\cdot(1-j)+(2+j)j$

$= 5-5j+4j-4j^2+2j+j^2$

$= 5+1j-3j^2$

$= 8+j$

c) $(1+2j)^2-(3+4j)$

$(a+b)^2 = (a+b)\cdot(a+b)$

$= a^2+2ab+b^2$

$1+\cancel{4j}+4j^2-3-\cancel{4j}$

$-6$

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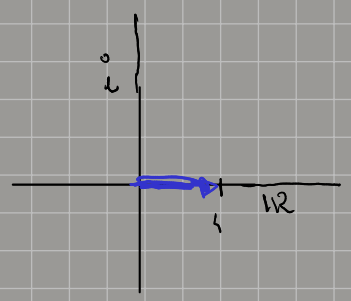
3) $|z| = \sqrt{a^2+b^2}$ , $\theta = \text{ARC}tg\left(\frac{b}{a}\right)$

a) $4$

$|z| = \sqrt{4^2+0} = 4$

$\theta = tg^{-1}\left(\frac{0}{4}\right)$

$z = 4\angle 0^\circ$

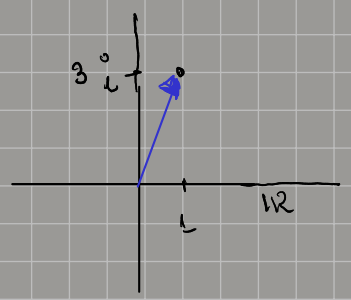



b) $1+\sqrt{3}j$

$|z| = \sqrt{1^2+\sqrt{3}^2} = \sqrt{1+3} = 2$

$\theta = tg^{-1}\left(\frac{\sqrt{3}}{1}\right) = 60^\circ$

$z = 2\angle 60^\circ$

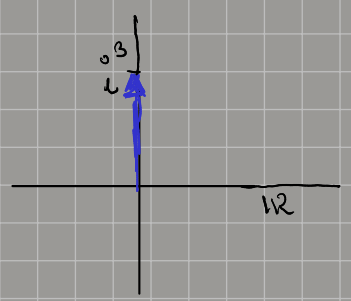



c) $3j$

$|z| = \sqrt{3^2} = 3$

$\theta = tg^{-1}\left(\frac{3}{0}\right) = 90^\circ$

$z = 3\angle 90^\circ$




d) $-\sqrt{2}+\sqrt{2}j$

$|z| = \sqrt{(-\sqrt{2})^2+\sqrt{2}^2}$

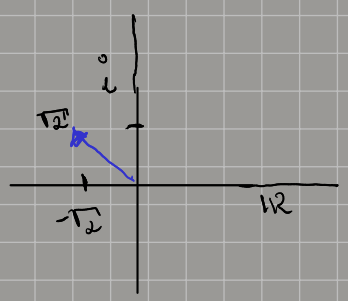
$|z| = \sqrt{2+2}$

$= \sqrt{4}$

$= 2$

$\theta = tg^{-1}\left(\frac{\sqrt{2}}{-\sqrt{2}}\right)$

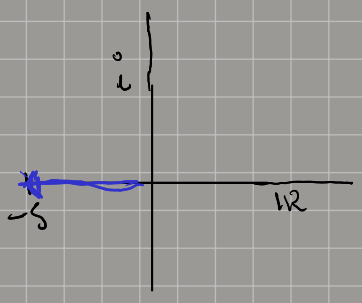
$\theta = -0{,}785$



e) $-5$

$|z| = \sqrt{(-5)^2 + 0^2} = \sqrt{25} = 5$

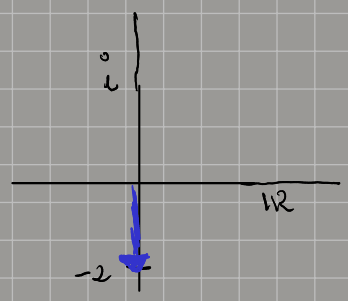
$\theta = tg^{-1}\left(\frac{0}{-5}\right) = 180^\circ$.




f) $-2j$

$|z| = \sqrt{0^2 + (-2)^2} = \sqrt{4} = 2$

$\theta = tg^{-1}\left(\frac{-2}{0}\right) = 270^\circ$

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4) a) $3 - 4j$

$|z| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

$\theta = tg^{-1}\left(\frac{-4}{3}\right) = -53{,}13^\circ$

b) $\sqrt{2} + \sqrt{2}j$

$|z| = \sqrt{(\sqrt{2})^2 + (\sqrt{2})^2} = \sqrt{4} = 2$

$\theta = tg^{-1}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = 45^\circ$

c) $12 + 5j$

$|z| = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$

$\theta = tg^{-1}\left(\frac{5}{12}\right) = 22{,}62^\circ$ $\quad z = 13 < 0{,}30{,}5^\circ$

d) $\cos(\theta) + j\,sen(\theta)$

$|z| = \sqrt{\cos^2 + sen^2} = \sqrt{1} = 1$

$\theta = \theta$

e) $-3+4j$

$|z| = \sqrt{(-3)^2+4^2} = \sqrt{25} = 5$

$\theta = Tg^{-1}\left(\frac{4}{-3}\right) = -53,13^\circ$ COMO ESTÁ NO 2° Q1 $\theta = -53,13 + 180$

$\theta = 126,87^\circ$

f) $7-2j$

$|z| = \sqrt{7^2+(-2)^2} = \sqrt{49+4} = \sqrt{53}$

$\theta = Tg^{-1}\left(\frac{-2}{7}\right) = -15,95^\circ$

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5) $z = |z| \cdot (\cos\theta + j\,\text{sen}(\theta))$

a) $3e^{j\pi}$

$\theta = 180^\circ$

$\cos 180^\circ = -1$

$\text{sen } 180^\circ = 0$

$|z| = 3$

$z = 3 \cdot (-1 + 0 \cdot j)$

$z = -3$

b) $4e^{j\frac{11\pi}{6}}$

$|z| = 4$

$\theta = \frac{11\pi}{6}$

$\cos\theta = 0,866$

$\text{sen } \theta = -0,5$

$z = 4(0,866 + (-0,5)j)$

$z = 3,464 - 2j$

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c) $2e^{j\pi/4}$

$\theta = \pi/4 = 45^\circ$

$|z| = 2$

$\cos\theta = \frac{\sqrt{2}}{2}$

$\text{sen}\,\theta = \frac{\sqrt{2}}{2}$

$z = \cancel{2} \cdot \left(\frac{\sqrt{2}}{\cancel{2}} + \frac{\sqrt{2}}{\cancel{2}} j\right)$

$z = \sqrt{2} + \sqrt{2}j$

d) $5e^{j3\pi/2}$

$\theta = \frac{3\pi}{2} = 270^\circ$

$|z| = 5$

$\cos\theta = 0$

$z = 5(0 + (-1)j)$

$\text{sen}\,\theta = -1$

$z = -5j$

e) $7e^{j\pi/2}$

$\theta = \frac{\pi}{2} = 90^\circ$

$|z| = 7$

$\cos\theta = 0$

$\operatorname{sen}\theta = 1$

$z = 7(0 + 1 \cdot j)$

$z = 7j$

f) $\theta = \pi/3 = 60^\circ$

$4e^{j\pi/3}$

$\cos = \frac{1}{2}$

$|z| = 4$

$\operatorname{sen}\theta = \frac{\sqrt{3}}{2}$

$z = 4\left(\frac{1}{2} + \frac{\sqrt{3}}{2}j\right)$

$z = 2 + 2\sqrt{3}j$

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6) a) $(1-j)(2+2j)$ $\quad j^2 = -1$

$2 + \cancel{2j} - \cancel{2j} - 2j^2$

$4$ $\quad |z| = \sqrt{4^2} = 4$

PARA RESOLVER ESSE EXERCÍCIO, FOI UTILIZADA A TÉCNICA DO △ DE PASCAL.

$(a+b)^6 = 1 \cdot a^6 b^0 + 6a^5b^1 + 15a^4b^2 + 20a^3b^3 + 15a^2b^4 + 6ab^5 + a^0b^6$

b) $(1 + \sqrt{3}j)^6$ $\quad j = \sqrt{-1},\ j^2 = -1,\ j^3 = -j,\ j^4 = 1,\ j^5 = i,\ j^6 = -1$ $\quad \sqrt{3}^6 = 3^{\frac{1}{2}\cdot 6} = 3^3 = 27$

$1^6 + 6 \cdot 1^5 \sqrt{3}j + 15 \cdot 1^4 \cdot (\sqrt{3}j)^2 + 20 \cdot 1^3 \cdot (\sqrt{3}j)^3 + 15 \cdot 1^2 \cdot (\sqrt{3}j)^4 + 6 \cdot 1 \cdot (\sqrt{3}j)^5 + 1 \cdot (\sqrt{3}j)^6$

$1 + 6\sqrt{3}j + 15 \cdot 3 \cdot j^2 + 20 \cdot 3 \cdot \sqrt{3} \cdot j^3 + 15 \cdot 9 \cdot j^4 + 6 \cdot 9\sqrt{3} \cdot j^5 + 27 \cdot j^6$

$1 + 6\sqrt{3}j + 45 \cdot (-1) + 60\sqrt{3} \cdot (-j) + 135 \cdot 1 + 54\sqrt{3} \cdot j + 27(-1)$

$1 - 45 + 135 - 27 + \cancel{6\sqrt{3}j} - \cancel{60\sqrt{3}j} + \cancel{54\sqrt{3}j}$

$z = 64$ $\quad |z| = \sqrt{64^2} = 64$

c) $\frac{3+3j}{1+2j} \cdot \frac{(1-2j)}{(1-2j)} = \frac{3 - 6j + 3j - \cancel{6j^2}^{-1}}{1^2 - 4j^2}$

$(a+b) \cdot (a-b)$

$a^2 - \cancel{ab} + \cancel{ab} - b^2$

$a^2 - b^2$

$z = \frac{9 - 3j}{5}$ $\quad |z| = \sqrt{\left(\frac{9}{5}\right)^2 + \left(\frac{-3}{5}\right)^2}$

$|z| = 1{,}897$

OBS: EU FIZ NA MÃO O ANTERIOR E ESQUECI DE USAR AS PROPRIEDADES.

ENTÃO USANDO ELAS:

a) $|z_1 \cdot z_2| = |z_1| \cdot |z_2| = \sqrt{2} \cdot \sqrt{8} = \sqrt{16} = 4$

$z_1 = (1-j)$, $|z_1| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$

$z_2 = (2+2j)$, $|z_2| = \sqrt{2^2 + 2^2} = \sqrt{8}$

b) $|z_1^x| = |z_1|^x \Rightarrow |z^6| = 2^6 = 64$

$z = 1 + \sqrt{3}j$

$|z| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$

c) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$ $\qquad \left|\frac{z_1}{z_2}\right| = \frac{\sqrt{18}}{\sqrt{5}} = \sqrt{\frac{18}{5}} \approx 1{,}897$

$z_1 = 3+3j$, $|z_1| = \sqrt{3^2 + 3^2} = \sqrt{18}$

$z_2 = 1+2j$, $|z_2| = \sqrt{1^2 + 2^2} = \sqrt{5}$

DEFINITIVAMENTE MAIS TRANQUILO :)

7) $\dfrac{1}{1-j} - \dfrac{1}{j}$

$$\frac{1\,j}{(1-j)\,j} - \frac{(1-j)}{(1-j)\,j}$$

$$\frac{j-1+j}{j-j^2} = \frac{2j-1}{j+1} \cdot \left(\frac{j-1}{j-1}\right)$$

$$= \frac{2j^2-2j-j+1}{j^2-1}$$

$$= -\left(\frac{-1-3j}{2}\right) = \frac{1+3j}{2}$$

$$|z| = \sqrt{\left(\frac{1}{2}\right)^2+\left(\frac{3}{2}\right)^2}$$

$$|z| = \sqrt{\frac{1}{4}+\frac{9}{4}}$$

$$|z| = \sqrt{\frac{10}{4}}$$

$$|z| = \frac{\sqrt{10}}{2}$$

$$\theta = Tg^{-1}\left(\frac{+3}{+1}\right) = 71{,}56$$

$$z = \frac{\sqrt{10}}{2} \angle 71{,}56^\circ$$

8) a) $2+5j$

$a = 2$

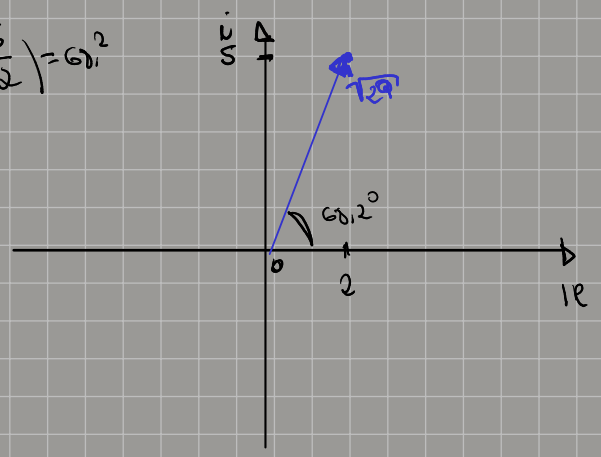
$b = 5$

$$|z| = \sqrt{2^2+5^2}$$

$$|z| = \sqrt{4+25}$$

$$|z| = \sqrt{29}$$

$$\theta = Tg^{-1}\left(\frac{5}{2}\right) = 68{,}2$$




b) $-3+2j$

$a = -3$

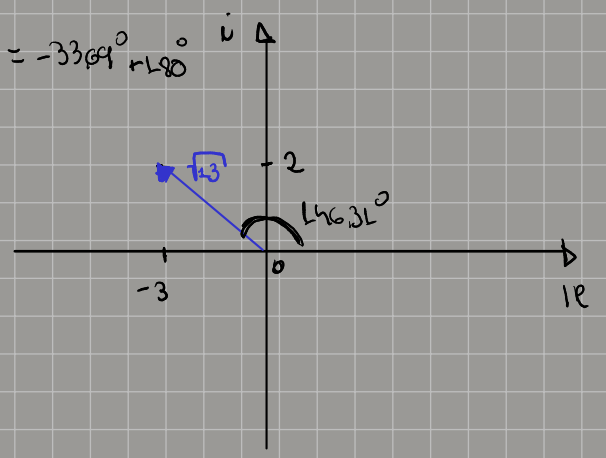
$b = 2$

$$|z| = \sqrt{(-3)^2+2^2}$$

$$= \sqrt{9+4}$$

$$= \sqrt{13}$$

$$\theta = Tg^{-1}\left(\frac{2}{-3}\right) = -33{,}69^\circ + 180^\circ$$



c) $-2-3j$

$\theta = tg^{-1}\left(\frac{-3}{-2}\right) = 56,31 + 180$

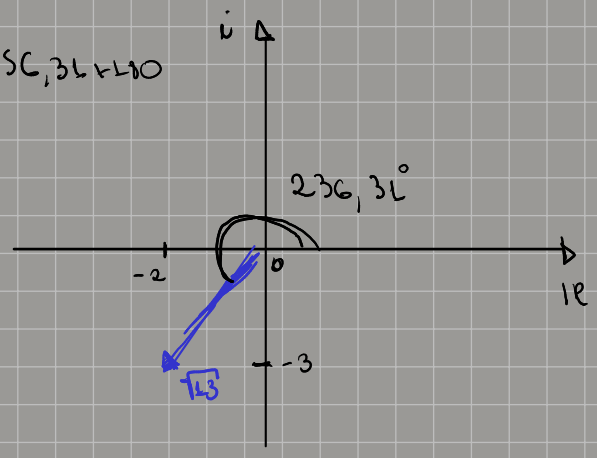
$a = -2$

$b = -3$

$|z| = \sqrt{(-2)^2 + (-3)^2}$

$|z| = \sqrt{4+9}$

$|z| = \sqrt{13}$



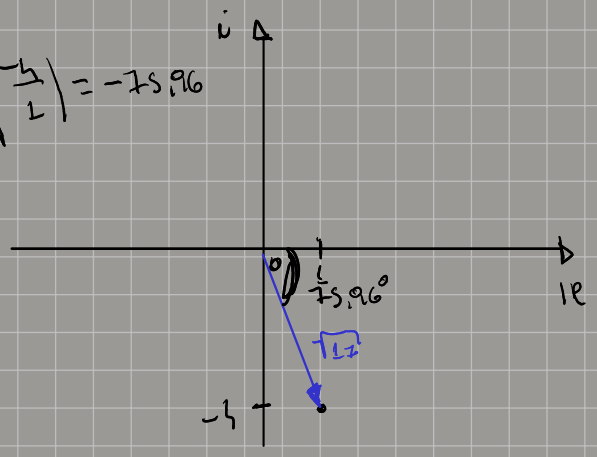

d) $1-4j$

$\theta = tg^{-1}\left(\frac{-4}{1}\right) = -75,96$

$a = 1$

$b = -4$

$|z| = \sqrt{1^2 + (-4)^2}$

$|z| = \sqrt{1+16}$

$|z| = \sqrt{17}$